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INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

MUSIC STORE MANAGEMENT

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AKATSUKI

MUSIC STORE

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INTRODUCTION

- The Music domain requires efficient handling of artists, albums, tracks, customers, employees, and sales transactions.
- Proper data management enables the store to track music sales, customer purchases, and revenue across different countries and cities.
- This project simulates a complete digital music store database with interconnected entities such as artists, genres, invoices, and playlists.
- It provides hands-on experience in understanding relational database design and real-world business workflows in the entertainment industry.
- We will use SQL queries to analyze customer behavior, popular genres, top sartists, and sales performance.
- The objective is to support data-driven decision-making and enhance operational and business insights within a music retail environment.

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TABLES

ALBUM TABLE

Field	Type	Null	Key	Default
album_id	int	NO	PRI	NULL
title	varchar(160)	YES		NULL
artist_id	int	YES	MUL	NULL

ARTIST TABLE

Field	Type	Null	Key	Default
artist_id	int	NO	PRI	NULL
name	varchar(120)	YES		NULL

ARTIST TABLE

Field	Type	Null	Key	Default
customer_id	int	NO	PRI	NULL
first_name	varchar(120)	YES		NULL
last_name	varchar(120)	YES		NULL
company	varchar(120)	YES		NULL
address	varchar(255)	YES		NULL
city	varchar(100)	YES		NULL
state	varchar(100)	YES		NULL
country	varchar(100)	YES		NULL
postal_code	varchar(20)	YES		NULL
phone	varchar(50)	YES		NULL
fax	varchar(50)	YES		NULL
email	varchar(100)	YES		NULL
support_re...	int	YES	MUL	NULL

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INVOICE TABLE

Field	Type	Null	Key	Default
invoice_id	int	NO	PRI	NULL
customer_id	int	YES	MUL	NULL
invoice_date	date	YES		NULL
billing_address	varchar(255)	YES		NULL
billing_city	varchar(100)	YES		NULL
billing_state	varchar(100)	YES		NULL
billing_country	varchar(100)	YES		NULL
billing_postal_code	varchar(20)	YES		NULL
total	decimal(10,2)	YES		NULL

EMPLOYEE TABLE

Field	Type	Null	Key	Default
employee_id	int	NO	PRI	NULL
last_name	varchar(120)	YES		NULL
first_name	varchar(120)	YES		NULL
title	varchar(120)	YES		NULL
reports_to	int	YES		NULL
levels	varchar(255)	YES		NULL
birthdate	date	YES		NULL
hire_date	date	YES		NULL
address	varchar(255)	YES		NULL
city	varchar(100)	YES		NULL
state	varchar(100)	YES		NULL
country	varchar(100)	YES		NULL
postal_code	varchar(20)	YES		NULL
phone	varchar(50)	YES		NULL
fax	varchar(50)	YES		NULL
email	varchar(100)	YES		NULL

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GENRE TABLE

Field	Type	Null	Key	Default
genre_id	int	NO	PRI	NULL
name	varchar(120)	YES		NULL

MEDIATYPE TABLE

Field	Type	Null	Key
media_type_id	int	NO	PRI
name	varchar(120)	YES	

INVOICE_LINE TABLE

Field	Type	Null	Key
invoice_line_id	int	NO	PRI
invoice_id	int	YES	MUL
track_id	int	YES	MUL
unit_price	decimal(10,2)	YES	
quantity	int	YES	

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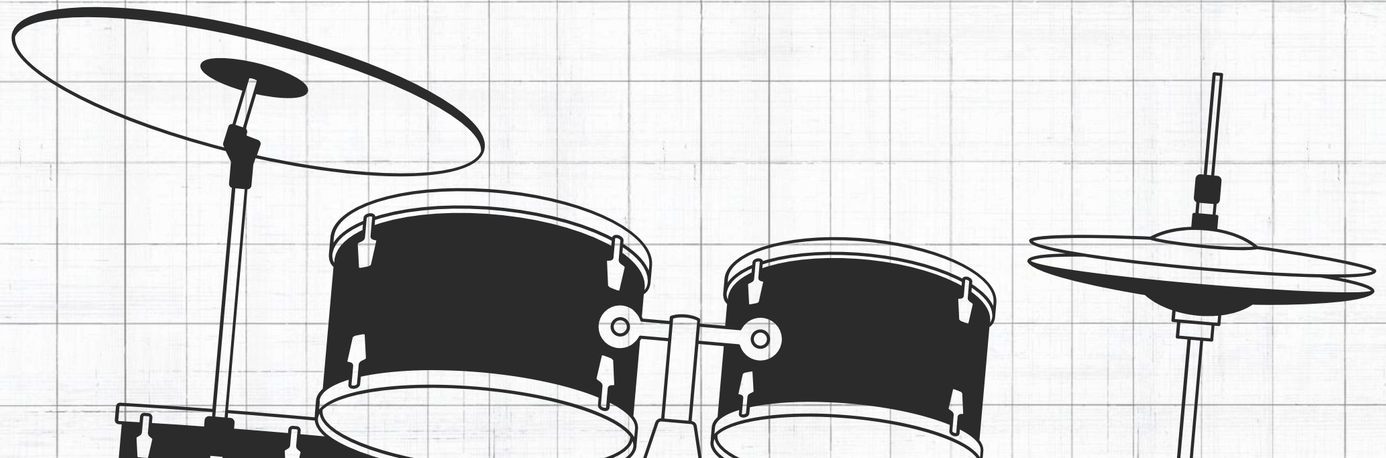
PLAYLIST TABLE

Field	Type	Null	Key	Default
playlist_id	int	NO	PRI	NULL
name	varchar(255)	YES		NULL

PLAYLIST_TRACK TABLE

Field	Type	Null	Key	Default
playlist_id	int	NO	PRI	NULL
track_id	int	NO	PRI	NULL



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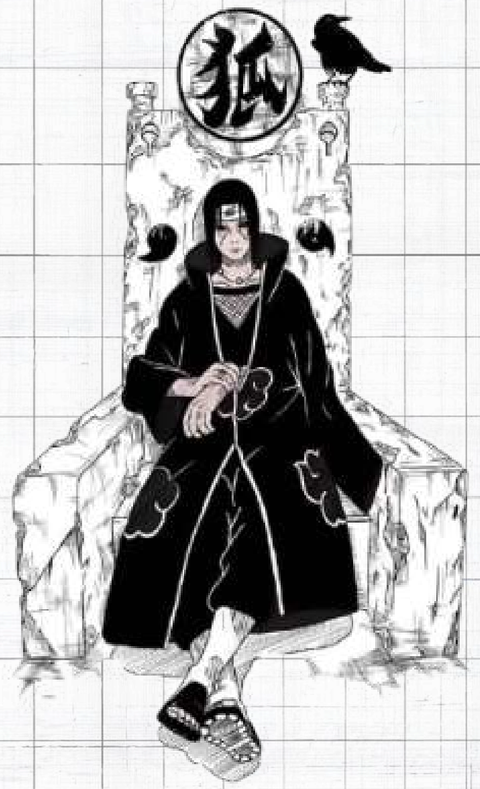
TRACK TABLE

Field	Type	Null	Key	Default
track_id	int	NO	PRI	NULL
name	varchar(200)	YES		NULL
album_id	int	YES	MUL	NULL
media_type_id	int	YES	MUL	NULL
genre_id	int	YES	MUL	NULL
composer	varchar(220)	YES		NULL
milliseconds	int	YES		NULL
bytes	int	YES		NULL
unit_price	decimal(10,2)	YES		NULL



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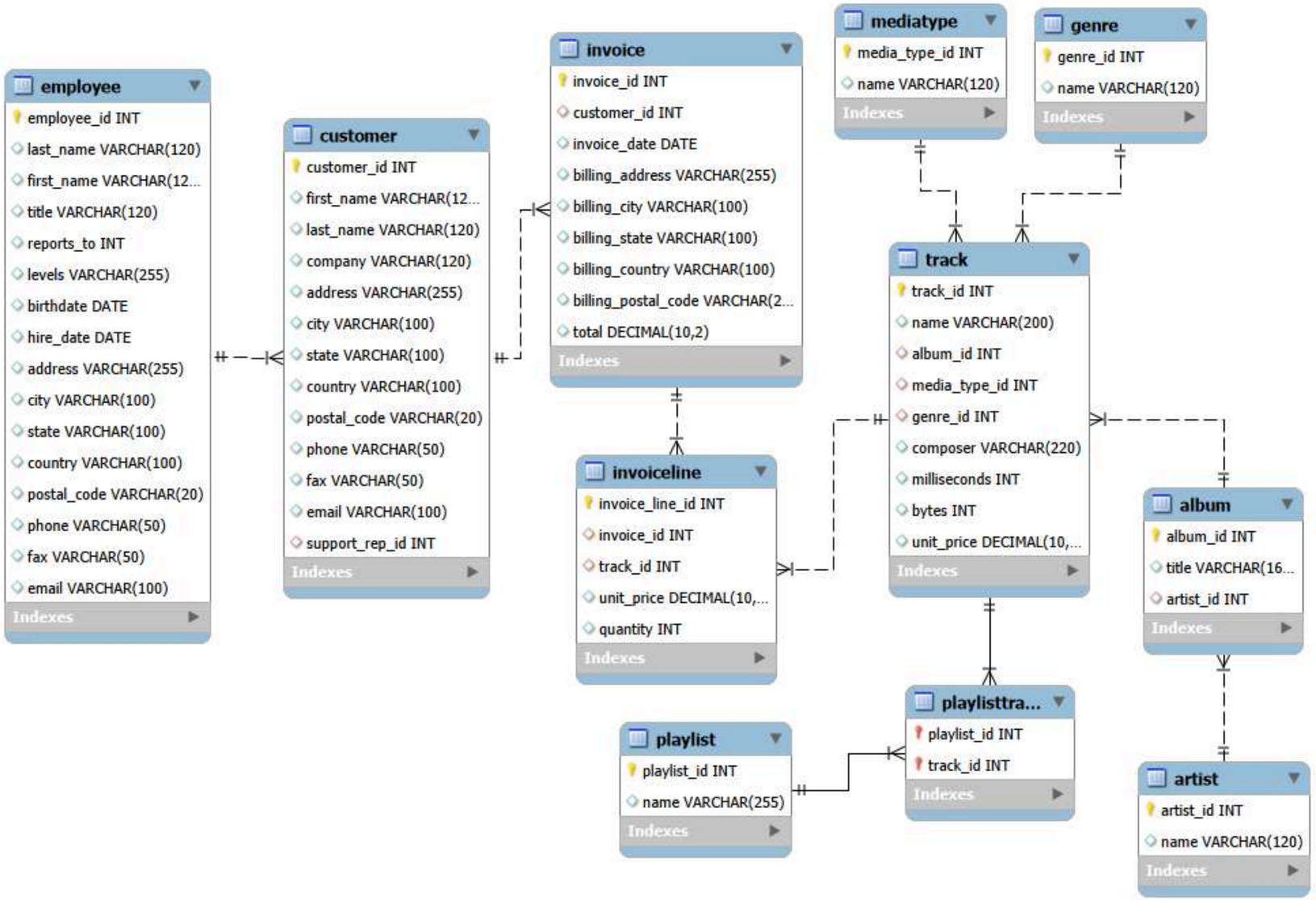
ER DIAGRAM



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1. WHO IS THE SENIOR MOST EMPLOYEE BASED ON JOB TITLE?

```
SELECT
* FROM Employee
ORDER BY hire_date
LIMIT 1;
```

employee_id	last_name	first_name	title
9	Madan	Mohan	Senior General Manager

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2. WHICH COUNTRIES HAVE THE MOST INVOICES?

```sql
SELECT billing_country,
COUNT(*) AS total_invoices
FROM Invoice
GROUP BY billing_country
ORDER BY total_invoices DESC;
```

billing_country	total_invoices
USA	131
Canada	76
Brazil	61
France	50
Germany	41
Czech Republic	30
Portugal	29
United Kingdom	28
India	21
Ireland	13

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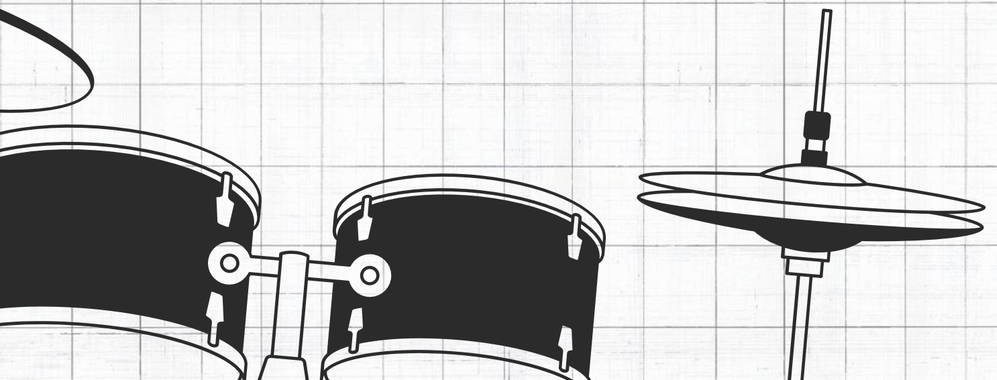
3. WHAT ARE THE TOP 3 VALUES OF TOTAL INVOICE?

```
SELECT
total from invoice
ORDER BY total
DESC LIMIT 3;
```

total
23.76
19.80
19.80



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4. WHICH CITY HAS THE BEST CUSTOMERS? - WE WOULD LIKE TO THROW A PROMOTIONAL MUSIC FESTIVAL IN THE CITY WE MADE THE MOST MONEY. WRITE A QUERY THAT RETURNS ONE CITY THAT HAS THE HIGHEST SUM OF INVOICE TOTALS. RETURN BOTH THE CITY NAME & SUM OF ALL INVOICE TOTALS

```
SELECT billing_city,
SUM(total)
AS Revenue from invoice
GROUP BY billing_city
ORDER BY Revenue DESC
LIMIT 1;
```

billing_city	Revenue
Prague	273.24

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5. WHO IS THE BEST CUSTOMER? - THE CUSTOMER WHO HAS SPENT THE MOST MONEY WILL BE DECLARED THE BEST CUSTOMER. WRITE A QUERY THAT RETURNS THE PERSON WHO HAS SPENT THE MOST MONEY

```sql
SELECT c.customer_id, c.first_name, c.last_name,
SUM(i.total) AS Total_Spent FROM customer c
JOIN invoice i ON c.customer_id = i.customer_id
GROUP BY c.customer_id, c.first_name, c.last_name
ORDER BY Total_Spent DESC
LIMIT 1;
```

customer_id	first_name	last_name	Total_Spent
5	František	Wichterlová	144.54

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6. WRITE A QUERY TO RETURN THE EMAIL, FIRST NAME, LAST NAME, & GENRE OF ALL ROCK MUSIC LISTENERS. RETURN YOUR LIST ORDERED ALPHABETICALLY BY EMAIL STARTING WITH A

```
SELECT c.email, c.first_name, c.last_name,
g.name AS Genre FROM customer c
JOIN invoice i ON c.customer_id = i.customer_id
JOIN invoiceline il ON i.invoice_id = il.invoice_id
JOIN track t ON il.track_id = t.track_id
JOIN genre g ON t.genre_id = g.genre_id
WHERE g.name = 'Rock'
ORDER BY c.email;
```

email	first_name	last_name	Genre

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7. LET'S INVITE THE ARTISTS WHO HAVE WRITTEN THE MOST ROCK MUSIC IN OUR DATASET. WRITE A QUERY THAT RETURNS THE ARTIST NAME AND TOTAL TRACK COUNT OF THE TOP 10 ROCK BANDS

```sql
SELECT a.name, COUNT(t.track_id)
AS Rock_Track_Count FROM artist a
JOIN album al ON a.artist_id = al.artist_id
JOIN track t ON al.album_id = t.album_id
JOIN genre g ON t.genre_id = g.genre_id
WHERE g.name = 'Rock'
GROUP BY a.artist_id, a.name
ORDER BY Rock_Track_Count DESC
LIMIT 10;
```

name	Rock_Track_Count

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8. RETURN ALL THE TRACK NAMES THAT HAVE A SONG LENGTH LONGER THAN THE AVERAGE SONG LENGTH.- RETURN THE NAME AND MILLISECONDS FOR EACH TRACK. ORDER BY THE SONG LENGTH, WITH THE LONGEST SONGS LISTED FIRST

```
SELECT name, milliseconds FROM track
WHERE milliseconds >
(SELECT AVG(milliseconds) FROM track)
ORDER BY milliseconds DESC;
```

name	milliseconds
Occupation / Precipice	5286953
Through a Looking Glass	5088838
Greetings from Earth, Pt. 1	2960293
The Man With Nine Lives	2956998
Battlestar Galactica, Pt. 2	2956081
Battlestar Galactica, Pt. 1	2952702
Murder On the Rising Star	2935894
Battlestar Galactica, Pt. 3	2927802
Take the Celestra	2927677
Fire In Space	2926593

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9. FIND HOW MUCH AMOUNT IS SPENT BY EACH CUSTOMER ON ARTISTS? WRITE A QUERY TO RETURN CUSTOMER NAME, ARTIST NAME AND TOTAL SPENT

```
SELECT CONCAT(c.first_name, " ", c.last_name)
AS Cutomer_Name, a.name AS Artist_Name,
SUM(il.unit_price * il.quantity)
AS Total_Spent FROM customer c
JOIN invoice i ON c.customer_id = i.customer_id
JOIN invoiceline il ON i.invoice_id = il.invoice_id
JOIN track t ON il.track_id = t.track_id
JOIN album al ON t.album_id = al.album_id
JOIN artist a ON al.artist_id = a.artist_id
GROUP BY c.customer_id, a.artist_id
ORDER BY total_spent DESC;
```

Cutomer_Name	Artist_Name	Total_Spent
Hugh O'Reilly	Queen	27.72
Wyatt Girard	Frank Sinatra	23.76
Helena Holý	Red Hot Chili Peppers	19.80
Aaron Mitchell	James Brown	19.80
Robert Brown	Creedence Clearwater Revival	19.80
François Tremblay	The Who	19.80
František Wichterlová	Kiss	19.80
Niklas Schröder	Queen	18.81
Hugh O'Reilly	Nirvana	18.81

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10. WE WANT TO FIND OUT THE MOST POPULAR MUSIC GENRE FOR EACH COUNTRY. WE DETERMINE THE MOST POPULAR GENRE AS THE GENRE WITH THE HIGHEST AMOUNT OF PURCHASES. WRITE A QUERY THAT RETURNS EACH COUNTRY ALONG WITH THE TOP GENRE. FOR COUNTRIES WHERE THE MAXIMUM NUMBER OF PURCHASES IS SHARED, RETURN ALL GENRES

```sql
SELECT country, genre, purchases
FROM ( SELECT i.billing_country AS country,
g.name AS genre, COUNT(*) AS purchases,
RANK() OVER (PARTITION BY i.billing_country
ORDER BY COUNT(*) DESC) AS rnk
FROM invoice i
JOIN invoiceLine il ON i.invoice_id = il.invoice_id
JOIN track t ON il.track_id = t.track_id
JOIN genre g ON t.genre_id = g.genre_id
GROUP BY country, genre) ranked
WHERE rnk = 1;
```

country	genre	purchases
Argentina	Alternative & Punk	17
Australia	Rock	34
Austria	Rock	40
Belgium	Rock	26
Brazil	Rock	205
Canada	Rock	333
Chile	Rock	61

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11. WRITE A QUERY THAT DETERMINES THE CUSTOMER THAT HAS SPENT THE MOST ON MUSIC FOR EACH COUNTRY. WRITE A QUERY THAT RETURNS THE COUNTRY ALONG WITH THE TOP CUSTOMER AND HOW MUCH THEY SPENT. FOR COUNTRIES WHERE THE TOP AMOUNT SPENT IS SHARED, PROVIDE ALL CUSTOMERS WHO SPENT THIS AMOUNT

```
SELECT cs.country, cs.first_name, cs.last_name, cs.total_spent
FROM ( SELECT i.billing_country AS country, c.customer_id,
c.first_name, c.last_name, SUM(i.total) AS total_spent,
RANK() OVER (PARTITION BY i.billing_country
ORDER BY SUM(i.total) DESC) AS rnk
FROM Customer c
JOIN Invoice i ON c.customer_id = i.customer_id
GROUP BY i.billing_country, c.customer_id, c.first_name, c.last_name) cs
WHERE rnk = 1
ORDER BY cs.country;
```

country	first_name	last_name	total_spent
Argentina	Diego	Gutiérrez	39.60
Australia	Mark	Taylor	81.18
Austria	Astrid	Gruber	69.30
Belgium	Daan	Peeters	60.39
Brazil	Luís	Gonçalves	108.90
Canada	François	Tremblay	99.99
Chile	Luis	Rojas	97.02
Czech Republic	František	Wichterlová	144.54
Denmark	Kara	Nielsen	37.62
Finland	Terhi	Hämäläinen	79.20
France	Wyatt	Girard	99.99
Germany	Fynn	Zimmermann	94.05
Hungary	Ladislav	Kovács	78.21

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THANK YOU...

GROUP MEMBERS:

- SATYANJANI GUBBALA
- ADITYA KUMAR LOTTI
- THEJASRI THALLAPALLI